

Kunal Maheshwari

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SUMMARY

OPT-EAD holder with full-time work authorization and over 4 years of experience as a Software Developer and Data Engineer, specializing in building scalable microservices and distributed data pipelines. Possesses a Master of Science in Computer Science with expertise in Java, Python, Spark, SQL, AWS, and Azure, complemented by strong CI/CD automation skills and graduate-level work on large-scale data platforms and privacy-preserving AI applications. Eager to leverage advanced skills in Machine Learning, Infrastructure as Code (IaC), and cloud-native technologies to excel as a Senior Software Engineer, Cloud and DevOps.

EDUCATION

Central Michigan University | Mount Pleasant, MI Aug 2024 - Dec 2025
Master of Science | Computer Science

Lovely Professional University | Punjab, India Aug 2016 - May 2020
Bachelor of Technology | Computer Science and Engineering

EXPERIENCE

Software Developer | NTT DATA | Bengaluru, India (Remote) August 2022 - August 2024

- Designed and optimized high-performance REST and SOAP microservices using Java 8/11, Spring Boot, and Azure Functions, successfully handling over 50,000 daily API requests with 99.99 percent SLA uptime.
- Engineered and automated robust CI/CD pipelines utilizing Jenkins, Maven, and Azure DevOps, facilitating continuous delivery across Azure Cloud and Oracle WebLogic environments, incorporating Tracing.
- Spearheaded modernization of legacy systems by migrating to Log4j 2.x and enhancing monitoring, reducing issue resolution time by 20 percent.
- Led a significant EDX platform upgrade, streamlining application architecture and dependency management to achieve 50 percent faster delivery cycles and improve Workflows.
- Collaborated within Agile and Scrum teams to deliver FHIR-compliant healthcare solutions using Git, SVN, and JUnit testing, implementing Performance Tuning practices.

Senior Analyst | Capgemini | Mumbai, India (Remote) October 2021 - August 2022

- Engineered robust Java and EJB-based enterprise applications and RESTful web services for major automotive clients, enhancing system performance by 90 percent through SQL query optimization and database tuning.
- Implemented efficient CI/CD pipelines using Jenkins and GitLab, integrating automated testing and quality gates for streamlined deployments.
- Mentored junior developers through comprehensive code reviews, debugging sessions, and promotion of clean coding practices, fostering knowledge sharing and skill development.

Analyst | Capgemini | Mumbai, India (Remote) July 2020 - October 2021

- Maintained high-availability enterprise systems with over 99 percent uptime through proactive monitoring and adaptive performance optimization strategies.
- Enhanced SQL data pipelines and database operations, improving query performance by 10 percent and significantly strengthening data integrity.
- Implemented comprehensive unit testing using JUnit4, achieving 99 percent code coverage and ensuring robust application stability.
- Supported release planning and sprint execution using JIRA within Agile development teams, contributing to efficient project management.

PROJECTS

NYU Taxi Data Analysis Platform (Spark, AWS, Terraform) January 2024 - May 2025

- Constructed a distributed Spark-based data pipeline to process and analyze over 8 GB of NYC taxi trip data in near real time, integrating Linux environments for seamless operations.
- Developed an interactive Flask web application for dynamic data exploration and visualization, enabling insightful analysis.
- Containerized ingestion, processing, and analytics components using Docker for reproducible deployments and Infrastructure as Code (IaC) principles with Terraform.
- Optimized Spark memory usage and partitioning strategies, reducing overall job runtime by 40 percent.
- Automated cloud infrastructure provisioning using Terraform across AWS services including S3, RDS, ECS, Lambda, and QuickSight, demonstrating proficiency in cloud-native development and IaC.

Privacy-Preserving AI Email Client (Capstone Project) (T5, LoRA) January 2025 - December 2025

- Engineered a fine-tuning pipeline for T5-based transformer models to perform advanced email summarization and classification, leveraging Machine Learning techniques.
- Applied parameter-efficient fine-tuning using LoRA to support low-resource training under stringent privacy constraints.
- Implemented model evaluation using ROUGE metrics and integrated quantization for secure local deployment, exploring Emerging Technologies in AI.
- Utilized advanced Python libraries for model development and experimented with advanced concepts in Networking for secure data handling.

SKILLS

- Programming Languages: Proficient in Java, Python, and SQL for comprehensive software development and data manipulation.

- Frameworks and Libraries: Expertise in Spring Boot, Spring Framework, Apache Spark, PySpark, Hugging Face, T5, LoRA, and JUnit for building scalable applications and advanced AI models.
- Cloud Platforms: Skilled in AWS (EC2, RDS, S3, Lambda) and Azure (Functions, DevOps, ServiceBus) for cloud-native development and Infrastructure as Code (IaC) implementation.
- Databases: Experienced with PostgreSQL, Oracle Database, IBM DB2, and RDS for robust data management and performance tuning.
- Tools and Technologies: Adept with Docker, Kubernetes, Jenkins, Git, SVN, Maven, JIRA, Confluence, Tableau, DBeaver, and Terraform for CI/CD, automation, and infrastructure management.
- Concepts: Strong understanding of ETL Pipelines, Data Warehousing, Data Analytics, Agile, Scrum, Test Driven Development, Object Oriented Programming, CI/CD, Microservices, Large-scale data platforms, Infrastructure automation, and Privacy-preserving LLM-based applications.
- Data Analysis and Visualization: Proficient in Tableau and QuickSight for insightful data analysis and dashboard creation.
- DevOps and Cloud: Extensive experience with CI/CD, IaC, Linux environments, Tracing, Workflows, Incident Response, and Capacity Planning.
- Machine Learning and Emerging Technologies: Applied knowledge in Machine Learning, GPU utilization, and exploring Emerging Technologies for innovative solutions.
- Performance and Systems: Focused on Performance Tuning, Networking, and ensuring robust system design.